Lawrence Melvin S

9789046500 | lawrence2000april@gamil.com | GitHub | LinkedIn | Chennai, Tamil Nadu

EXPERIENCE

Wind River Mar. 2026 – Present
Engineer Sys Chennai, Tamil Nadu

- Contributed to migrating service-to-service communication from **Kafka to HTTP**, simplifying inter-service communication and improving integration flexibility.
- Designed and implemented a resilience strategy using **circuit breakers, configurable retries, fallback thresholds, exponential backoff, and jitter** to prevent retry storms and cascading failures.
- Applied circuit breakers selectively to **GET and non-database-dependent operations**, while allowing POST operations to bypass circuit breaking unless explicitly configured, preserving transactional behavior.
- Implemented **on-demand agent pod provisioning** with a short cooldown mechanism, enabling dynamic resource allocation and improving infrastructure utilization.
- Resolved production defects involving **hardware information not being captured in Redis** and process configuration issues; performed root-cause analysis and documented corrective procedures to prevent recurrence.

BNP Paribas Aug. 2022 – Mar. 2026
Software Engineer Chennai, Tamil Nadu

- Designed Python automation for monthly **non-regression testing**, reducing manual QA effort by approximately **40%** while dynamically integrating new Spring Batch tasks.
- Integrated financial data from **Amazon S3 and third-party KYC APIs** into risk reports using Python.
- Developed **Spring Batch jobs** for bulk file processing and ingestion of **1M+ records**, ensuring timely generation of trade and financial reports.
- Developed risk-analysis screens using **Spring and ExtJS**, improving data visualization for front-office teams.
- Upgraded **Spring dependencies** across multiple repositories, improving application stability and resolving security vulnerabilities.

BNP Paribas Jan. 2022 – Aug. 2022
Software Engineer Intern Chennai, Tamil Nadu

- Identified and resolved application vulnerabilities using **Fortify**, reducing security risks across enterprise applications.
- Increased **SonarQube code coverage by 50%** through comprehensive JUnit test implementation.
- Reconciled Ubix and market data using **WTX and custom SQL scripts**, improving financial data consistency.
- Managed **WAR file generation and deployment using WebLogic** for enterprise Java applications.

PROJECTS

InsureTracks | React, Spring Boot, PostgreSQL

- Built a full-stack insurance tracking application using **React, Spring Boot, and PostgreSQL** with Google OAuth2 authentication and automated policy-expiry reminders.
- Designed REST APIs for policy management and integrated the React frontend with a Spring Boot backend.

EDUCATION

Loyola-ICAM College of Engineering and Technology (LICET) Jul. 2018 – May 2022
Bachelor of Engineering in Computer Science Chennai, Tamil Nadu

SKILLS

Languages : Python, Java, Perl, SQL (Oracle)
Backend : Spring Boot, Spring Batch, Spring, Flask, Django
Frontend : React, ExtJS
Databases : Oracle, PostgreSQL
Cloud & DevOps : AWS, Docker, Jenkins, Git
Testing & Tools : JUnit, SonarQube, Fortify, Postman
AI/ML : Scikit-Learn, Pandas, Hugging Face, LangChain